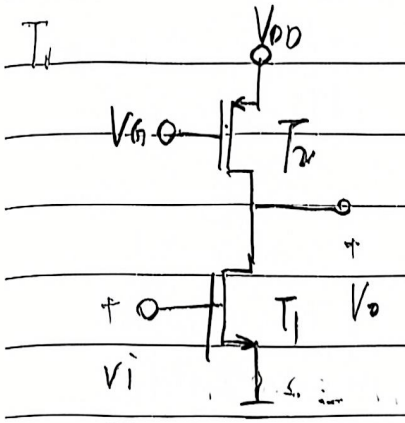
No

Date.

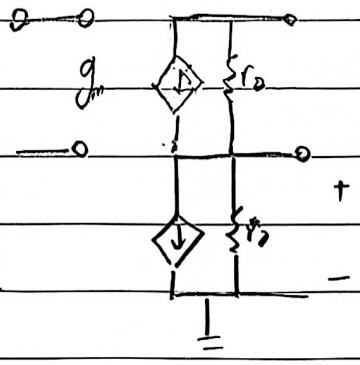
Ch3



$(\frac{W}{L})_1 = 1 \quad (\frac{W}{L})_2 = \frac{1}{4}$

$\mu_n C_{ox} = 2\,mA/V^2$

$V_A = 50V, \ I_D = 200\mu A$



$r_o = \frac{V_A}{I_D} = 250k\Omega$

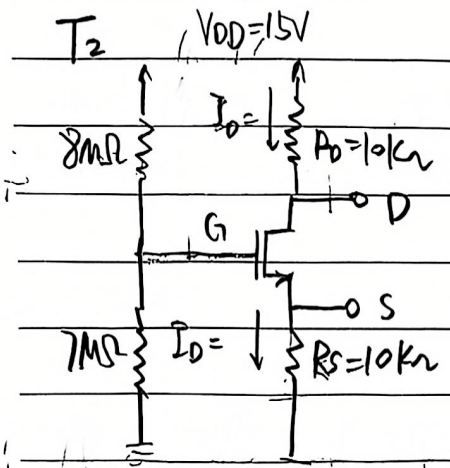
$R_o = \frac{r_o}{2} = 125k\Omega$

$g_{m2} = \mu C_{ox}(\frac{W}{L})_1 (V_{GS} - V_t)$

$I_D = \frac{1}{2}\mu C_{ox}(\frac{W}{L})_1 (V_{GS} - V_t)^2$

$\Rightarrow g_{m2} = \sqrt{2 I_D \mu C_{ox}(\frac{W}{L})_1} = 8.94 \times 10^{-5}\Omega^{-1}$

$A_v = \frac{V_o}{V_i} = -g_m R_o = -11.18$



$V_t = 1V, \ \frac{W}{L} = 5, \ \mu_n C_{ox} = 0.2mA/V^2$ 求 I_D

$V_{GS} = 7 - 10 I_D$

$V_{DS} = 15 - 20 I_D$

若处于饱和区

$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_t)^2 \Rightarrow I_D = 0.5mA$ 或 $0.7mA$ (舍)

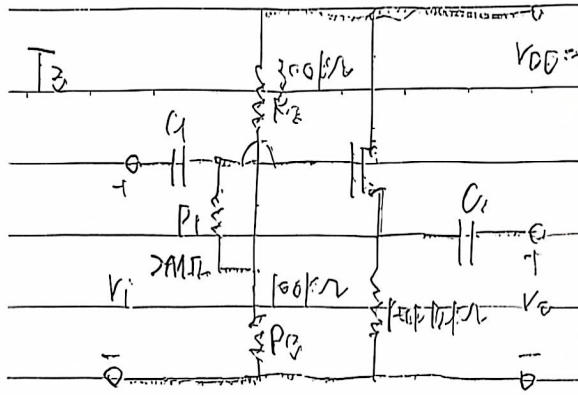
此时 $V_{DS} = 5V, \ V_{GS} = 2V$

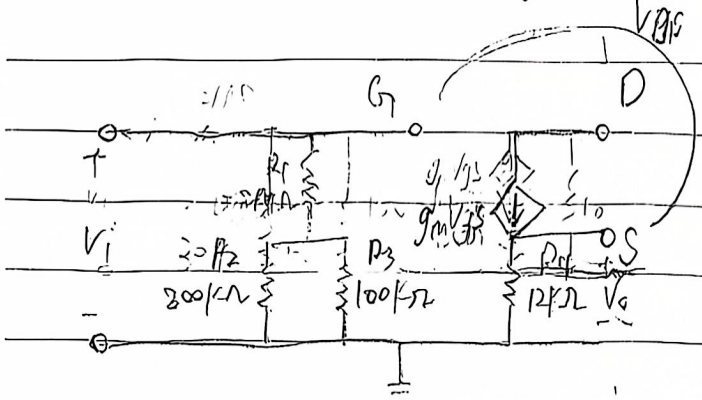
符合饱和区条件

No.

Date.

T3



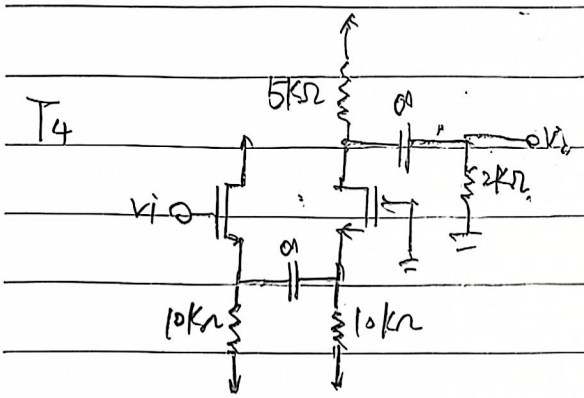


$R_i = R_1 + R_2 // R_3 = 2.075M\Omega$

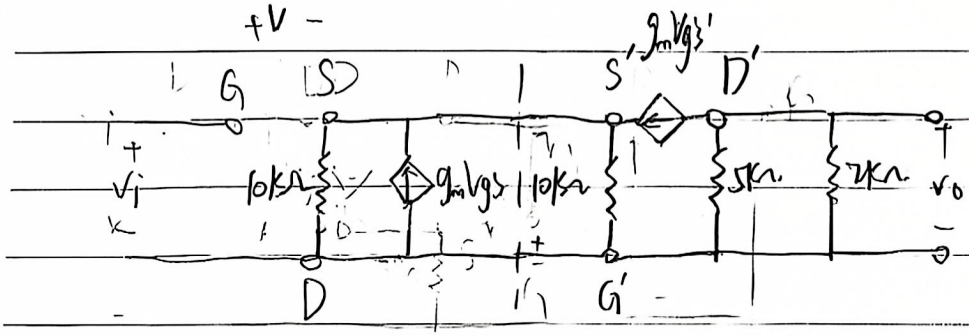
$R_o = \frac{V}{g_mV + \frac{V}{R_4}} = 1.017K\Omega$

$A_M = \frac{g_mV_{gs}gR_4}{V_{gs}+g_mV_{gs}R_4} = 0.915$

T4



$g_m = 1mA/V$，求 V_o/V_i



$R_{i2} = \frac{V}{g_mV + \frac{V}{10}} = 0.099K\Omega$

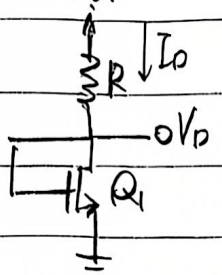
$A_{v1} = \frac{-g_mV(10//R_{i2})}{g_mV(10//R_{i2})+V} = 0.495$ $\quad \frac{V_o}{V_i} = A_{v1}A_{v2} = 7.07$

$A_{v2} = \frac{g_mV(5//2)}{V} = -14.284$

No.

Date

T5



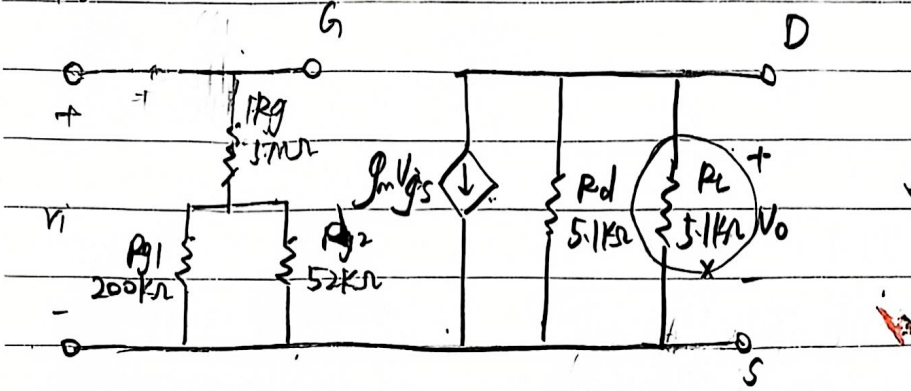
$V_{tn}=0.5V$ $\mu_n C_{ox}=0.4mA/V^2$ $\frac{W}{L}=4$ 要求 $V_D=0.7V$ 求 R

$V_{DD}=V_{GS}$ $V_{DD}>V_{GS}-V_t$，工作在饱和区

$\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_D-V_{tn})^2R+V_D=1.8$

$\Rightarrow R=34.375K\Omega$

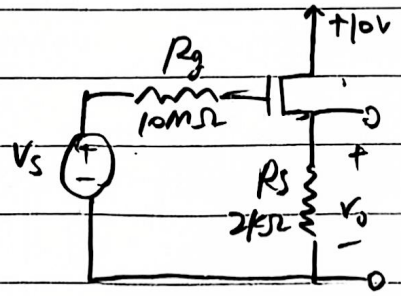
T6

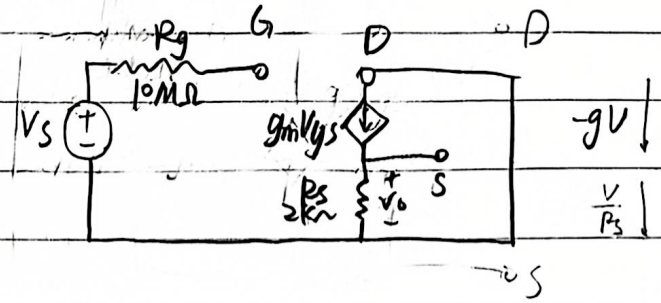


$g_m=2mS$

(a) $A_v=\frac{-g_mV_{gs}(R_d/\!/R_L)}{V_{gs}}=-5.1$ $R_o=R_d=5.1K\Omega$

$R_i=R_g+(R_{g1}/\!/R_{g2})=504.1K\Omega$





(b) $A_v=\frac{g_mV_{gs}R_S}{V_{gs}+g_mV_{gs}R_S}=0.8$ $R_o=\frac{V}{g_mV+\frac{V}{R_S}}=0.4K\Omega$

$R_i=\infty$

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